

GNAPing On the Job: Attacking and Defending Facial Detection on Edge Devices

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Abstract—Facial detection systems, particularly those deployed on edge devices, have become crucial components of modern security infrastructures. However, their susceptibility to adversarial attacks raises significant concerns, particularly when attackers exploit signal-dependent noise to target critical features like facial landmarks. This paper introduces the Guided-inspired Noise Attack Pyramid (GNAP), a novel adversarial attack model that leverages the Laplacian-of-Gaussian (LoG) and Laplacian Pyramid filters to reduce CNN-based facial recognition accuracy by targeting high-frequency regions of images. We evaluate GNAP's effectiveness through experiments on the Labeled Faces in the Wild (LFW) dataset and real-time video processing on edge devices, demonstrating up to an 18 percent reduction in classification confidence with minimal computational impact. To counter GNAP, we propose the Guided Noise Attack Guard (GNAG) defense strategy, utilizing an Unsharp Mask filter to restore classification confidence to near-original levels without sacrificing real-time performance. The combined insights offer a balanced approach to both attacking and defending facial recognition systems on resource-constrained edge devices. The code to reproduce our results is available at <https://github.com/ChiefAj23/GNAPing-On-the-Job.git>.

Index Terms—Adversarial Attacks, Edge Devices, Convolutional Neural Networks, Laplacian-of-Gaussian (LoG), Laplacian Pyramid Filter, Guided-inspired Noise Attack Pyramid (GNAP), Guided Noise Attack Guard (GNAG)

I. INTRODUCTION

MACHINE LEARNING (ML) has revolutionized security systems and has been extensively adopted by governments, businesses, and individuals to safeguard sensitive information [2]. Among various ML techniques, **Convolutional Neural Networks (CNNs)** have advanced facial recognition technologies for robust authentication and access control mechanism [3]. Increasingly, these technologies are being deployed on **edge devices**, which perform local processing instead of relying on cloud-based servers. Edge computing offers low latency and scalability benefits, making it ideal for real-time facial detection in areas

such as smartphone security and smart city surveillance [2] [4].

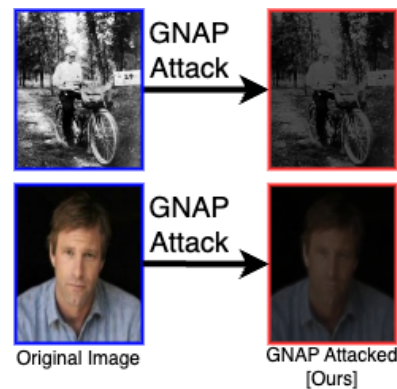


Fig. 1. **Attack Overview:** This figure illustrates the GNAP attack applied to a random image.

However, edge devices face significant challenges due to limited computational power and security infrastructures, making them vulnerable to adversarial attacks [5] [4]. Among these attacks, the **CiPer** method [6], which leverages the **Laplacian of Gaussian (LoG)** technique to introduce perturbations, is of particular concern. While defenses like **perturbation-aware filtering** have mitigated such attacks for static images [7], their effectiveness in real-time settings on resource-constrained devices remains limited. Other techniques, such as histogram equalization (HE), are demonstrated for particularly rain-affected or foggy weather and require high-end FPGAs to achieve real-time performance [8].

In this work, we investigate the **Laplacian Pyramid** technique [9], widely used for noise reduction and feature enhancement, and explore how adversarial attacks can combine LoG and Laplacian Pyramid methods to strengthen their impact on facial detection systems. To address this, we propose a novel defense that integrates **adaptive filtering** and multiresolution techniques to counter these attacks in real-time, providing a practical and efficient solution. Overall, this paper first investigates a novel adversarial attack technique for edge devices, and then we address its solutions. This led to the

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following two research questions:

- **RQ1:** How does combining LoG and Laplacian Pyramid enhance adversarial attack success rates, and is executing these attacks in real-time on resource-constrained devices feasible?
- **RQ2:** Can we design a defense framework for edge devices to filter adversarially perturbed images from these techniques?

In summary, the novel contributions of this paper are:

- 1) **GNAP Attack (see Section II):** We introduce the **Guided Noise Attack Pyramid (GNAP)**, a novel adversarial attack combining LoG [6] and Laplacian Pyramid filters [9] to target facial landmarks and degrade recognition accuracy on edge devices.
- 2) **GNAG Defense (see Section III):** We propose the **Guided Noise Attack Guard (GNAG)**, a defense mechanism that employs real-time image enhancement techniques to mitigate GNAP's impact and ensure robust facial detection.

II. GNAP DISRUPTION: A NOVEL ADVERSARIAL ATTACK

This section introduces the Guide-inspired Noise Attack Pyramid (GNAP) model. We begin by discussing adversarial attacks against facial detection systems. Next, we provide explanation on our threat model. Finally, we conclude this section with details of the GNAP model's architecture and effectiveness.

A. Motivation

Adversarial Attacks on Facial Recognition Despite their widespread implementations, facial detection systems are vulnerable to adversarial attacks that leverage subtle perturbations on images to deceive CNNs [6] [2]. Traditional deception techniques often use signal-independent noise generation, like Gaussian noise, which is easily detectable and less representative of real-world experiences [6], [10], [11].

Limitations of Signal-Independent Noise Signal-independent noise applies statistically derived perturbations, which do not exploit the image's structure, particularly around crucial areas like facial landmarks [6]. Perturbations generated in this fashion are deterministic and thus easier to detect and neutralize, limiting the attack's impact [6].

Motivation for Signal-Dependent Approaches To address these limitations of Signal-Independent strategies, signal-dependent noise generation technique is investigated. This technique bases its perturbations on the image's intrinsic noise, which can be leveraged to target critical regions such as facial landmarks [6]. Inspired by the philosophy of utilizing intrinsic noise, we developed the GNAP model, which integrates signal-dependent techniques (details discussed in Section II-C) for a more robust attack on facial detection systems. Figure 1 shows overview of the effectiveness of the attack on the image. Using the GNAP model, the

attacker disrupts facial recognition by exploiting the image's intrinsic noise. The model's efficiency and low computation requirements make it suitable for deployment on edge devices.

B. Threat Model

Our threat model assumes that the adversary has intermediate access between the camera and CNN, which enables the attacker to intercept and modify the images before they are passed to CNN, as seen in Figure 2. The threat model reflects a real-world scenario where attacker may exploit communication channels/device firmware vulnerabilities. This process needs minimal resources and brief access to initiate subtle attacks.

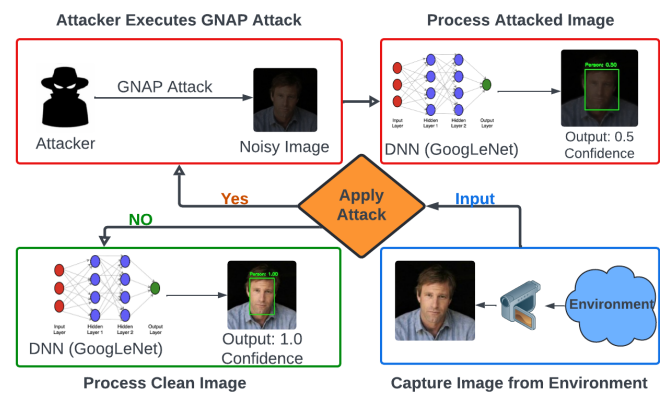


Fig. 2. Threat Model: The attacker intercepts the image post-capture and implements GNAP to reduce classification confidence.

C. Explanation of the Proposed GNAP Attack

The GNAP model combines the Laplacian-of-Gaussian (LoG) filter [6] and the Laplacian Pyramid filter [9] to exploit intrinsic noise in input images. The GNAP model, showcased in Figure 3, targets facial landmarks, making it ideal for attacking facial detection systems. This section explores how these filters work in tandem with impact on CNN performance.

1) Laplacian-of-Gaussian (LoG) Technique: The first phase of GNAP, the Laplacian-of-Gaussian (LoG) [6] technique, applies a Gaussian blur to smooth low-frequency sections while preserving the edges of the image. The Laplacian filter then highlights these edges, concentrating noise around critical landmarks. This noise profile is subtracted from the original image. It weakens edge integrity, which is crucial for accurate facial recognition.

2) Laplacian Pyramid Technique: We introduce a Laplacian Pyramid [9] to amplify the LoG attack, which adds multi-scale noise. It down-scales the image to various resolutions and reconstructs them into a composite image. Finally, it injects subtle perturbations at each scale. This

results in a perturbed image with noise distributed across different scales, which makes the attack more resilient to traditional denoising strategies.

3) Design of the Proposed GNAP Attack: The GNAP model stands out by integrating signal-dependent techniques and focusing on high-frequency areas (landmarks) critical for facial recognition. Figure 3 shows that the camera output is perturbed first using LoG attack [6], and its output is fed to Laplacian Pyramid [9] filters. GNAP creates a more effective attack with minimal resource consumption, which is ideal for deployment on edge devices. Experimental results indicate (discussed in section IV-A3) that GNAP significantly reduces CNN classification confidence compared with individual attacks.

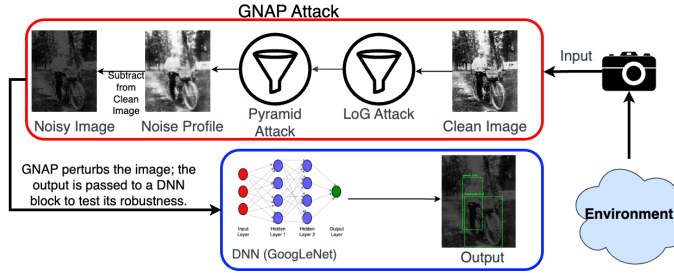


Fig. 3. GNAP Model in practice: The image is processed through the LoG and Laplacian Pyramid filters, generating a multi-scale noisy image that disrupts CNN classification.

D. GNAP Implementation and Mathematical Formulation

As alluded earlier, the GNAP technique synthesizes the Laplacian of Gaussian (LoG) filter with the Laplacian Pyramid filter to generate a targeted adversarial attack. The LoG filter operates by smoothing the image and isolating high-frequency components, which are crucial for edge detection. The Laplacian Pyramid filter then deconstructs the image into multiple scales, enabling the generation of noise profiles added across different image resolutions.

Mathematical Formulation

a) LoG Filter: The LoG filter is mathematically defined in [6] as follows:

$$\text{LoG}(x, y) = \frac{-1}{\pi\sigma^4} \left[1 - \frac{x^2 + y^2}{2\sigma^2} \right] \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

Where x and y denote the spatial coordinates, and σ represents the standard deviation of the Gaussian function, controlling the filter's sensitivity to different frequency components.

b) Laplacian Pyramid Filter: The Laplacian Pyramid is constructed as follows:

Gaussian Pyramid Generation:

$$G_i = \text{Downsample}(\text{GaussianFilter}(G_{i-1})) \quad (2)$$

Laplacian Pyramid Generation:

$$L_i = G_i - \text{Upsample}(\text{GaussianFilter}(G_{i+1})) \quad (3)$$

G_0 corresponds to the original image, and L_i represents the Laplacian image at level i . (Here, i is an index that refers to the level of the pyramid. The lower value of i corresponds to higher-resolution images and a higher value of i corresponds to lower-resolution images).

III. GNAG: A ROBUST NOISE DEFENSE STRATEGY

The Guide-inspired Noise Attack Pyramid (GNAP) is notably robust against conventional defense strategies, namely adversarial training.

Adversarial training has long stood as a cornerstone in defending against adversarial attacks [12]. However, when deployed on edge devices, this method presents substantial challenges. Adversarial training typically involves integrating adversarial examples into the model's learning process, a technique that, while comprehensive, imposes significant computational demands [13]. These demands are particularly problematic in resource-constrained environments such as edge devices, where performance is paramount. Furthermore, when confronted with multiple, potentially conflicting attacks, the overhead introduced by adversarial training can severely degrade system performance, compromising the responsiveness edge devices deliver.

To develop a robust defense mechanism without disrupting the underlying Convolutional Neural Network (CNN) model, we introduce a novel approach that preprocesses images using specialized defense filters before passing them to CNN. Our method ensures that if an unperturbed image is processed, the defense mechanism does not degrade the image quality or affect the model's confidence score. However, when the image is subjected to our robust GNAP attack, the defense mechanism effectively mitigates adversarial perturbations, resulting in a significantly improved confidence score compared to the attacked image.

The proposed GNAG (Guided Noise Attack Guard) defense mechanism uses the Unsharp Mask technique [14], a well-established image enhancement process, to enhance image resilience against adversarial noise by sharpening edges. The Unsharp Mask process consists of two primary steps as shown in Figure 4:

Step 1: Gaussian Blur

Gaussian Blur is a widely used image processing technique that smooths an image by reducing noise and detail through pixel averaging within a neighborhood defined by a Gaussian distribution. This blur is applied via a convolution operation, with the two-dimensional Gaussian function defined as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (4)$$

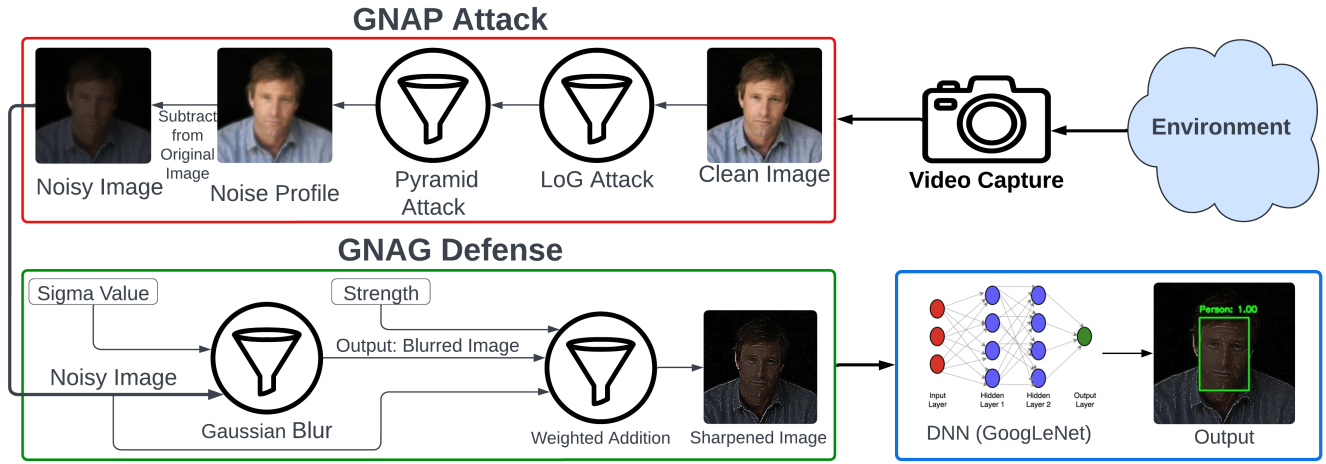


Fig. 4. GNAG-Defense: Illustration of the proposed defense mechanism.

where x and y represent the distances from the origin (center of the Gaussian kernel), and σ is the standard deviation, controlling the extent of the blur.

Step 2: Weighted Addition

Weighted addition combines the original and blurred images by assigning weights to each, thereby controlling the sharpening effect. The sharpened image is obtained using the following formula:

$$\text{Sharpened Image} = (1 + \text{str}) \times \text{OI} - \text{str} \times \text{BI} \quad (5)$$

where:

- **OI - Original Image:** The initial image intended for sharpening.
- **BI - Blurred Image:** The output of the Gaussian Blur applied to the Original Image.
- **str - strength:** A parameter determining the degree of sharpening applied.

Determining an effective strength value for sharpening is challenging due to noise variations. Our experiments indicate that a noise scaling factor of **0.7** requires a strength value of approximately **14**. This was determined through an iterative process, adjusting the strength until no further improvement in confidence was observed. We identified a linear relationship between noise scaling (**0.00 to 1.00**) and strength was noted, while the sigma value (controlling blur) had no significant impact on the results.

IV. RESULTS AND DISCUSSION

To rigorously assess the effectiveness of the Guided-inspired Noise Attack Pyramid (GNAP) model and Guided Noise Attack Guard GNAG, we implemented the attack in two distinct scenarios. The first scenario involved static image analysis using an LFW dataset [1]. The second scenario involved a real-time attack processed on a resource-constrained device.

This section explains the experiment setup, dataset, models, scenarios, and results of these experiments. Next, it discusses the experimental setup and results in relation to the proposed defense mechanism (GNAG)

A. GNAP Attack

1) *Experiment Setup:* To demonstrate the generalizability and robustness of the GNAP attack, we utilized a 1080p webcam in conjunction with a Raspberry Pi 5 as the edge device for static image and live feed scenarios, as illustrated in Figure 5. This setup simulates real-world deployment in resource-constrained environments. For facial detection, we employed the pre-trained GoogleNet [3], which is a well-established Convolutional Neural Network (CNN) model. The GNAP attack is applied to static images and live video streams, and performance metrics are recorded.

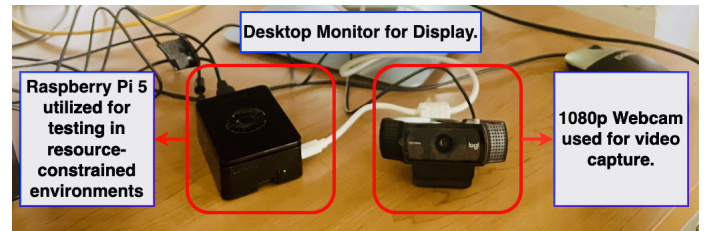


Fig. 5. Experiment Setup: This includes Raspberry PI 5 with 1080p webcam as edge device. The CNN model is deployed on Raspberry PI 5.

2) *Dataset:* Our experiments utilized the Labeled Faces in the Wild (LFW) dataset [1], a standard benchmark in facial detection research. LFW comprises over 13,000 images of more than 5,000 individuals. This dataset is particularly challenging due to its high variability in facial expressions, lighting conditions, and obfuscations.

3) *Scenarios and Results:* We examined the GNAP model on our experiment setup, as shown in Figure 5 across two distinct scenarios to evaluate its effectiveness in reducing the classification confidence of CNN models.

- **Scenario 1: Static Image Analysis** In this scenario, we applied the GNAP model to 100 randomly selected images from the LFW dataset [1] for evaluation. To compare the GNAP model's effectiveness with other filters, we independently passed 100 images through the Log filter [6], the Laplacian Pyramid filter [9], and the GNAP model. Each image before and after applying these filters and the GNAP attack is subsequently classified using the pre-trained GoogleNet [3] CNN model implemented in Caffe, an opensource deep learning framework optimized for speed and efficiency in large-scale image classification task [15]. The result is recorded in Table I. Table I shows the results, where the GNAP model led to an **18% average reduction in classification confidence** ($p < 0.05$), significantly higher than the negligible impact observed with the individual filters. Here, the p-value represents the statistical significance of the difference in classification confidence between the GNAP model and other filters. This result underscores the importance of the GNAP model (combining LoG and Laplacian Pyramid filter) in enhancing the effectiveness of the adversarial attack.

TABLE I
ATTACK EFFECTIVENESS ON LIVE FACES IN THE WILD(LFW) DATASET
AT 0.7 NOISE SCALE (CAFFE [15])

Attack Status	Mean Highest Confidence
Original	0.99
LoG [6]	0.97
Laplacian Pyramid [9]	0.99
GNAP Model [Ours]	0.82

- **Scenario 2: Real-time Video Analysis** We extended our evaluation to a real-time setting, implementing the GNAP model on a live video feed processed by a Raspberry Pi 5. Consistent with Scenario 1, the individual filters had minimal impact on CNN performance. However, the GNAP model caused a substantial drop in classification confidence. Results verify the findings from the static image analysis explored in scenario 1, as shown in Table II.

Moreover, the attack's impact on system performance was minimal. The average frame rate remained stable across all conditions (**mean FPS $\approx$ 2.83**), demonstrating the model's efficiency for real-time applications on edge devices.

B. GNAG Defense

1) *Scenario and Result:* We examined the GNAG defense model on our experiment setup, as shown in Figure 5 across

TABLE II
ATTACK EFFECTIVENESS ON LIVE FEED AT 0.7 NOISE SCALE (CAFFE [15])

Attack Status	Average Confidence	Average FPS
LoG [6]	1.00	2.58
Laplacian Pyramid [9]	1.00	3.01
GNAP Model [Ours]	0.80	2.89

two distinct scenarios. The primary objective is to evaluate its effectiveness in restoring classification confidence on GNAP attacked images when processed by CNN models.

- **Static Image Analysis:** In this scenario, we applied the GNAP attack to 100 randomly selected images from the LFW dataset [1] and processed them through the GNAG defense. Each image, having undergone both the GNAP attack and GNAG defense, was classified using the pre-trained GoogleNet CNN model [3]. As shown in Table III, GNAG nearly restored the classification confidence from **0.82** (post GNAP attack) to **0.98**, indicating its high effectiveness in mitigating adversarial perturbations.

TABLE III
GNAG EFFECTIVENESS ON LFW DATASET (CAFFE [15])

Attack Status	Mean Highest Confidence
Original	0.99
LoG [6]	0.97
Laplacian Pyramid [9]	0.99
GNAP Model [Ours]	0.82
GNAG Defense [Ours]	0.98

- **Real-time Video Analysis:** In the real-time setting, live video input from a 1080p webcam was processed by the GNAP attack on a Raspberry Pi 5, followed by the GNAG defense. The results in Table IV show the efficacy of GNAG defense effectiveness. The GNAP attack reduces classification confidence from **0.99** to **0.82**, but the GNAG defense nearly restores it to **0.98**, demonstrating significant recovery.

TABLE IV
GNAG EFFECTIVENESS ON LIVE FEED AT 0.7 NOISE SCALE (CAFFE [15]): DEMONSTRATES THE EFFECTIVENESS OF THE DEFENSE, WITH A NEAR-COMPLETE RESTORATION OF CONFIDENCE. NOTABLY, THE AVERAGE FPS REMAINED ROUGHLY THE SAME, INDICATING A LOW IMPACT ON RESOURCE USAGE.

Attack Status	Average Confidence	Average FPS
LoG [6]	1.00	2.58
Laplacian Pyramid [9]	1.00	3.01
GNAP Model [Ours]	0.80	2.89
GNAG Defense [Ours]	0.99	2.76

V. KEY INSIGHTS

This research on the GNAP attack and GNAG defense mechanisms provides key insights that address our research

questions and contribute to advancing adversarial attack and defense strategies in real-time, resource-constrained environments.

- **RQ1:** Integrating Laplacian-of-Gaussian (LoG) and Laplacian Pyramid in the GNAP model considerably boosts adversarial assault effectiveness by selectively perturbing high-frequency regions, particularly face landmarks. The combined approach leads to a **20%** loss in CNN classification confidence in real-time settings, surpassing the efficacy of individual filter-based attacks. GNAP was successfully implemented in real-time on a resource-constrained Raspberry Pi 5, displaying negligible computational overhead while maintaining constant performance (**mean FPS** $\approx$ 2.83). Results in Table II demonstrate the attack's practical feasibility on edge devices and expose its real-world security consequences against traditional defenses under limited settings.
- **RQ2:** The GNAG defense utilizes the Unsharp Mask filter and effectively neutralizes the GNAP attack by restoring CNN classification confidence accuracy to near-original levels across static and real-time scenarios. Its ability to counter adversarial perturbations with minimal computational overhead and no degradation in processing speed establishes it as a highly efficient defense mechanism against GNAP, even in performance-critical environments. The GNAG defense was also efficiently deployed in real-time on a Raspberry Pi 5, restoring classification confidence without impacting system performance. Results in Table IV demonstrate GNAG's robust applicability for real-time edge-device deployments, offering a practical solution to safeguarding facial detection systems without compromising operational efficiency.

VI. CONCLUSION

In this work, we propose **GNAP**: a novel adversarial attack targeting facial recognition systems on edge devices by leveraging the combination of Laplacian-of-Gaussian [6] and Laplacian Pyramid filters [9] to perturb high-frequency regions like facial landmarks. GNAP results in an **18%** reduction in classification confidence on static images and **20%** on real-time video while maintaining low computational overhead, making it suitable for deployment on resource-constrained systems. To counter this, we propose **GNAG**: a defense mechanism utilizing the Unsharp Mask filter to restore confidence to near-original levels. GNAG proves effective in both static and real-time scenarios, mitigating adversarial effects without compromising system performance. These findings emphasize the critical need for balancing offensive and defensive strategies to protect facial recognition technologies on edge devices, highlighting the importance of ongoing research in adversarial attacks and defenses.

VII. ACKNOWLEDGMENT

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